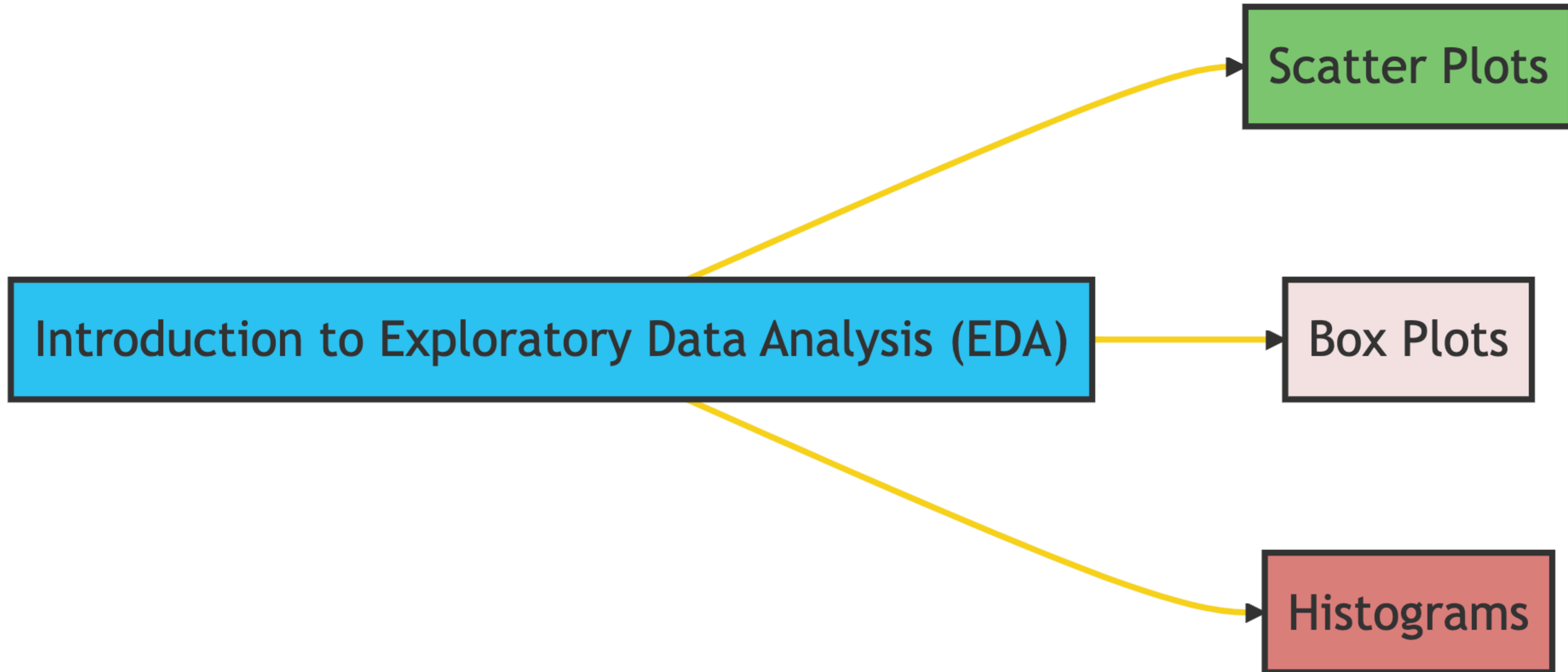


Exploratory Data Analysis

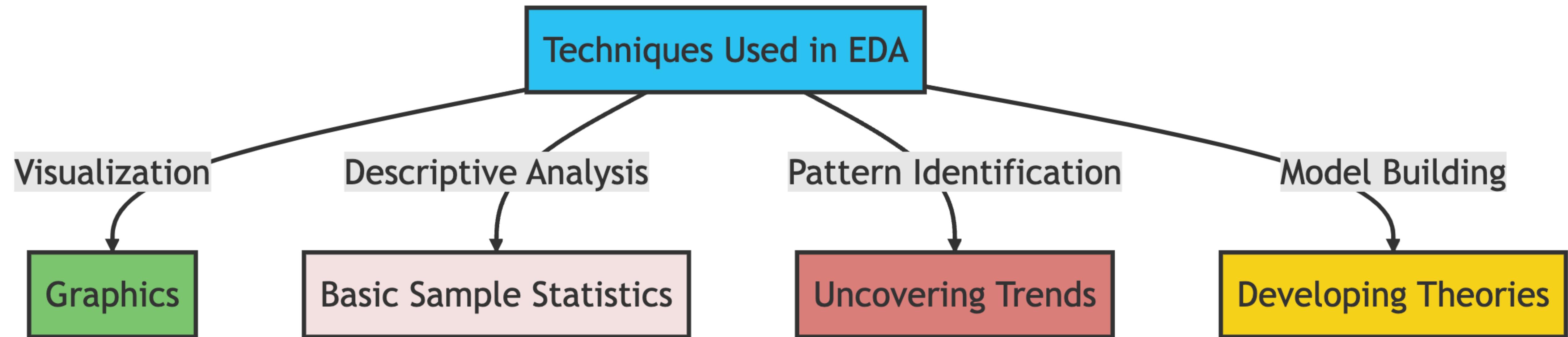
Introduction to Exploratory Data Analysis (EDA)

- 1. Definition of EDA**
- 2. Purpose of EDA**
- 3. Visual methods in EDA**
- 4. Comparison with other data analysis approaches**



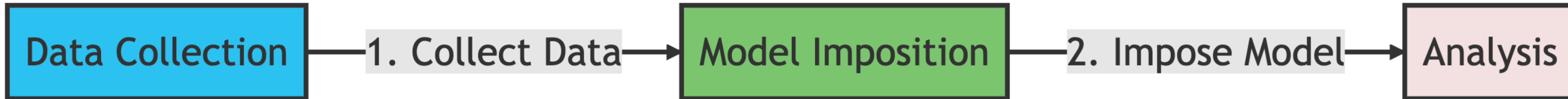
Techniques Used in EDA

1. Graphics
2. Basic sample statistics
3. Uncovering trends, outliers, and patterns
4. Developing theories for modeling



Classical Data Analysis

1. Sequence of intermediate steps
2. Imposition of a model
3. Focus on parameters of the model



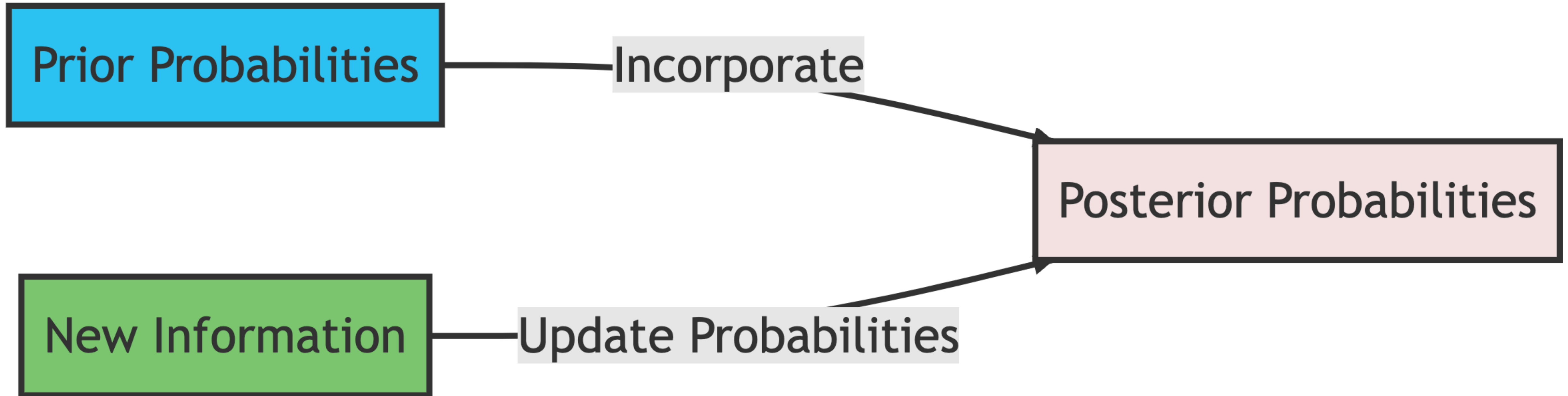
Exploratory Data Analysis (EDA) Approach

- 1. No initial model imposition**
- 2. Inference of appropriate models**
- 3. Data-driven model suggestions**



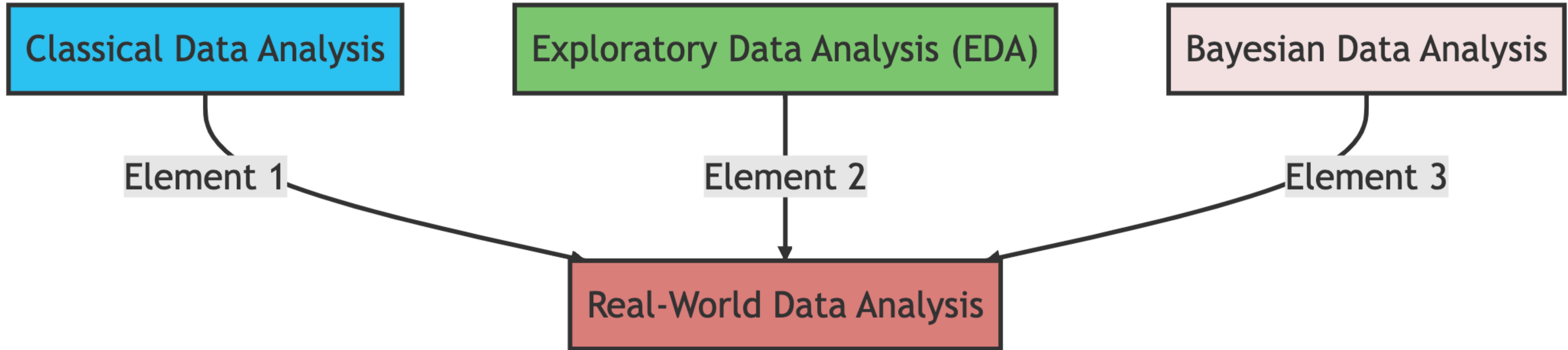
Bayesian Data Analysis

1. Research questions about unknown parameters
2. Use of prior data and domain knowledge
3. Purpose of determining posterior probabilities



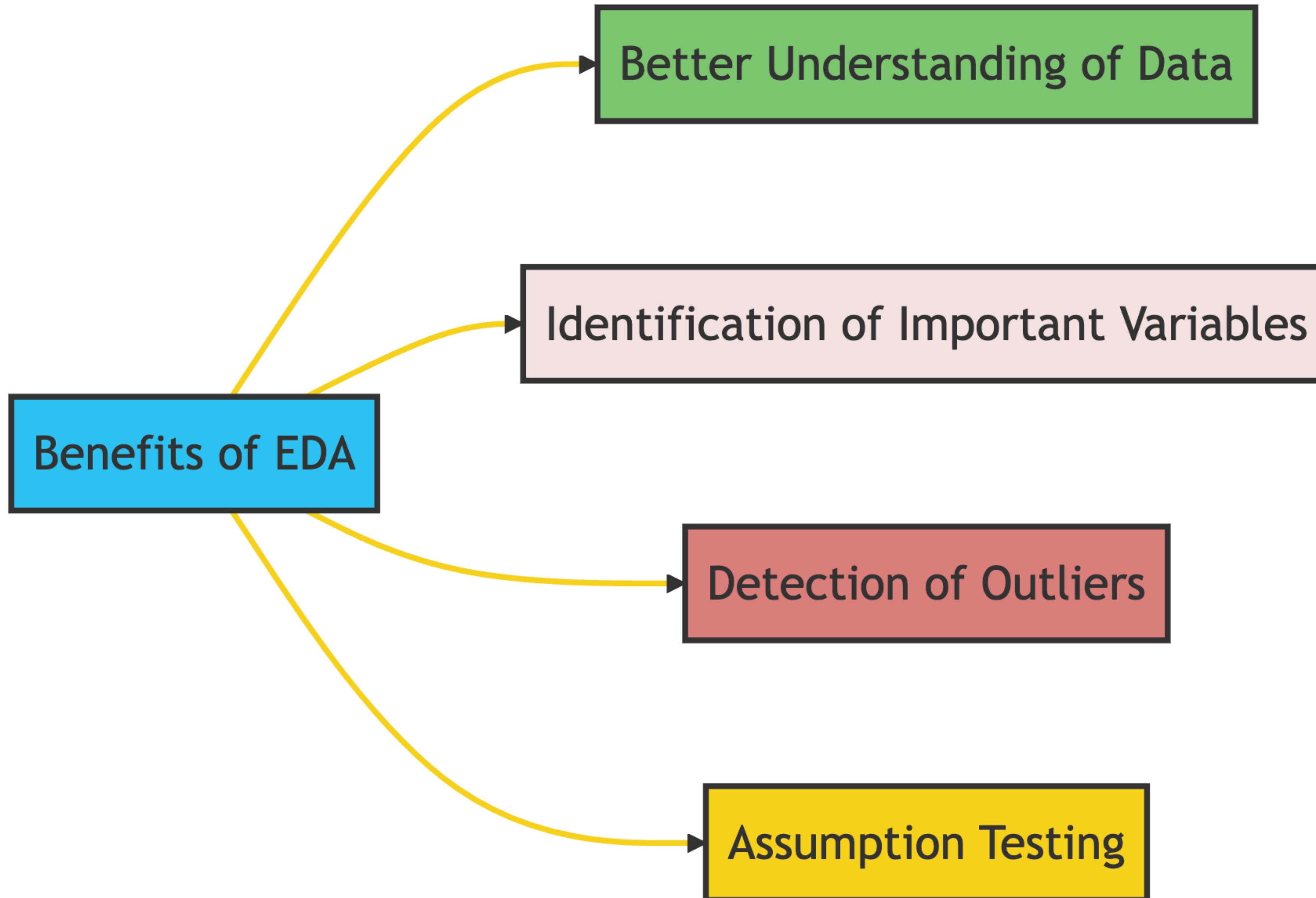
Mixing Approaches in Real-World Data Analysis

- 1. Combination of classical, EDA, and Bayesian approaches**
- 2. Use of other data analysis approaches**
- 3. Emphasis on major differences among the three approaches**



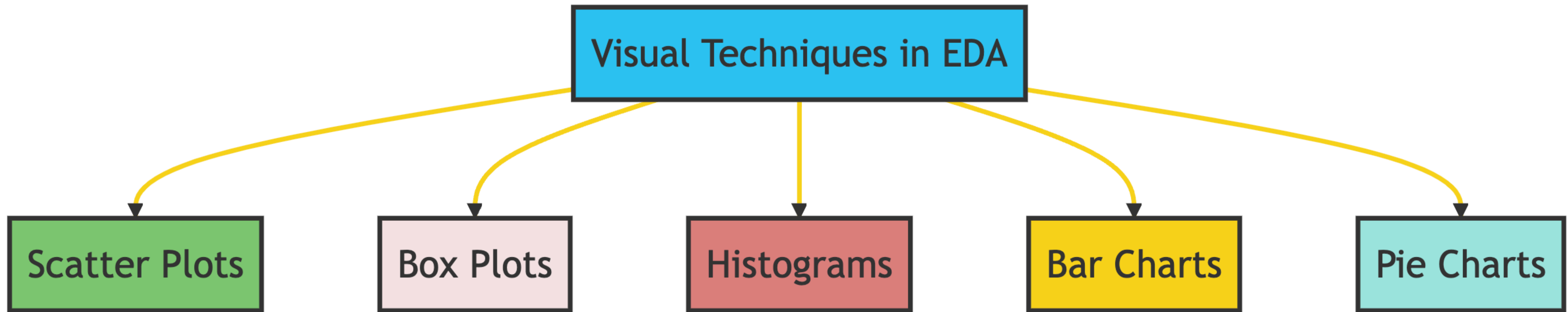
Benefits of EDA

1. Better understanding of data structure
2. Identification of important variables
3. Detection of outliers and anomalies
4. Assumption testing and model development



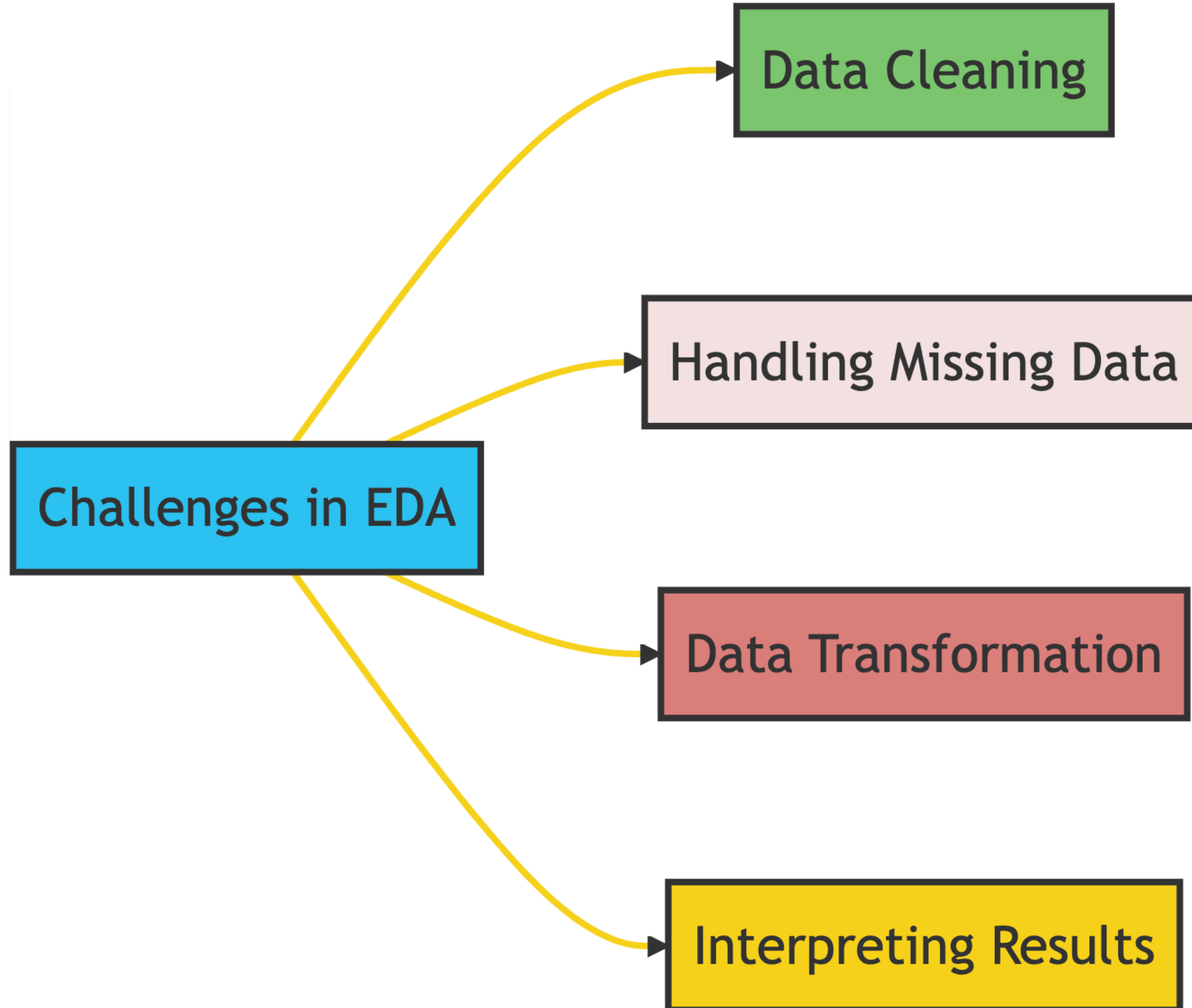
Visual Techniques in EDA

1. Scatter plots
2. Box plots
3. Histograms
4. Other graphical representations



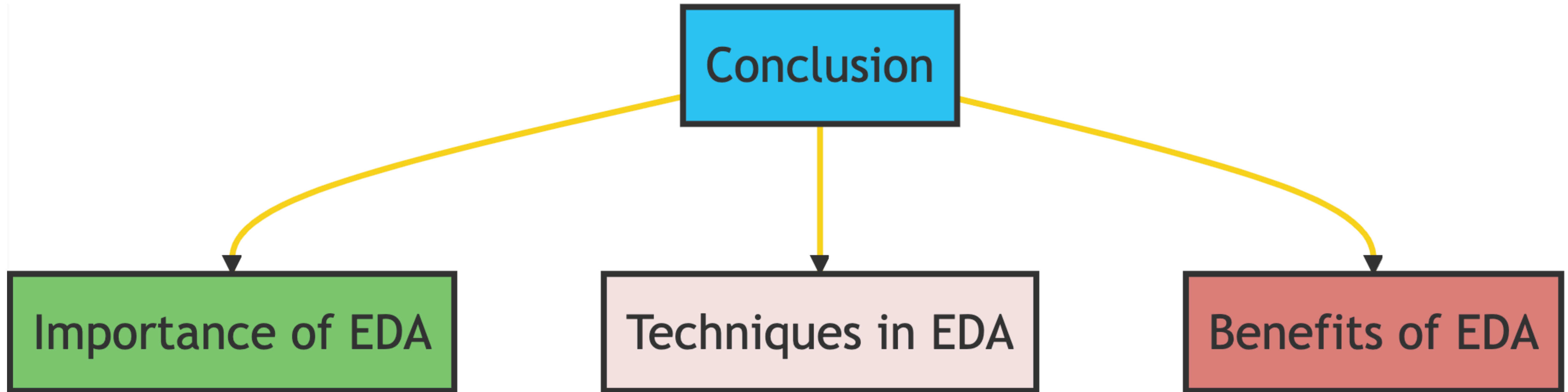
Challenges in EDA

1. Data cleaning
2. Handling missing data
3. Data transformation
4. Interpreting results



Conclusion

1. Importance of EDA in data analysis
2. Variety of techniques and approaches
3. Benefits of using EDA for informed decision-making



**Thanks
For
Watching**